MEA CO2 Capture Ver 1 · operator training simulator

ELECTRICAL LOAD LIST

Load list, demand, transformer / MCC sizing, cable schedule, hazardous area

MEA CO2 capture unit: absorber T-101 / stripper T-102 (30 wt% MEA)

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Teaching document generated for the MEA CO2 capture operator training simulator. Not for construction. Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data.

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Body pages are A4 landscape. Every number is read from the simulator model (design point, noise off) or marked assumed; the same data file (ll.json) feeds the Electrical tab of the OTS, which recomputes the checks from the live model.

1 · Basis and assumptions

Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data.

Item	Value	Source
LV system	400 V, 3-phase, 50 Hz	assumed 400/230 V, 3-phase, 50 Hz (Thai LV practice)
MV supply to K-101	6.6 kV	assumed 6.6 kV host-plant board for K-101
Design ambient	40 °C	assumed 40 °C design ambient
Motor rating rule	× 1.25 / 1.15 / 1.10	driver rating ≥ shaft power × 1.25 (< 22 kW), 1.15 (22–55 kW), 1.10 (> 55 kW): typical API 610 practice
Motor efficiency	IE3	IE3 4-pole minimum efficiency, IEC 60034-30-1 (typical)
Demand factors	C 1.0 · I 0.3 · S 0	demand factors: continuous 1.0, intermittent 0.3, standby 0 (typical load-list convention)
Transformer margin	× 1.25	assumed (spare capacity and growth)
Transformer impedance	4.0 %	assumed transformer impedance 4 % (IEC 60076-5 minimum up to 630 kVA)
Cable ampacity	table	Cu/XLPE multicore, method E, 30 °C (after IEC 60364-5-52 Table B.52.12; verify with the cable maker)
Derating	ambient 0.91 × grouping 0.8	XLPE in air at 40 °C (IEC 60364-5-52 Table B.52.14, typical); assumed grouping on ladder tray
Voltage-drop limits	running 5.0 % · DOL start 15.0 %	typical plant practice
DOL starting	6.5 × FLC at pf 0.3	typical
K-101 shaft power	313.1 kW	model JI-101 at design (176.8 kNm³/h, 80 % speed); the model includes a fan efficiency of 0.80
K-101 at 100 % speed	589 kW at 223 kNm³/h	model run, FIC-101 in MAN at 100 % (tools/run_model.js)

Pump hydraulic power

$P_{hyd} = Q \cdot \Delta P$, $\Delta P = (p_2 - p_1) + \rho g \Delta z + \Delta P_{exchanger} + \Delta P_{valve} + \Delta P_{line}$, $P_{shaft} = P_{hyd} / \eta_{pump}$, $P_{motor} \geq P_{shaft} \times$ margin → next IEC standard rating

Pump	Flow t/h (source)	ρ kg/m³	Q m³/h	p1 → p2 kPa a	Δz m (ρgΔz kPa)	ΔP exch / valve / line kPa	ΔP kPa (head m)	η pump	P hyd kW	P shaft kW
P-101	282.2 (FI109)	1112	253.7	107.0 → 180.9	13 (142)	70 / 70 / 30	386 (35.4)	0.72	27.19	37.76
P-102	269.7 (FIC201)	1020	264.4	182.4 → 103.0	24 (240)	120 / 70 / 40	391 (39.1)	0.72	28.70	39.86
P-103	5.5 (FI303)	992	5.6	175.0 → 180.9	18 (175)	0 / 70 / 30	281 (28.9)	0.40	0.44	1.09
P-104	297.6 (assumed)	992	300.0	103.0 → 103.0	5 (49)	70 / 0 / 60	179 (18.4)	0.70	14.89	21.27

Flows and densities are the model design point (FI-109, FIC-201, FI-303; rhoL at the sump conditions). Elevations, equipment and line losses and pump efficiencies are assumed (TBC by the pump hydraulic calculation). P-104: assumed 300 m³/h: E-105 5.8 MW (sheet) at ≈ 17 K rise; no wash flow in the model.

Worked example: P-101

$Q = 282.2$ t/h ÷ 1112 kg/m³ = 253.7 m³/h; $\Delta P = (180.9 - 107.0) + 142 + 70 + 70 + 30 = 386$ kPa

$P_{hyd} = 253.7/3600 \times 386 = 27.19$ kW; $P_{shaft} = 27.19/0.72 = 37.76$ kW; × 1.15 = 43.4 kW → **45 kW** motor

2 · Load list

Tag	Service	Duty	V	Start	Shaft kW	Motor kW	η motor	pf	Input kW	Demand kW	Demand kvar	Assumed
K-101	Flue gas blower, VFD (FIC-101 → speed)	continuous	6600	VFD	313.1	355	0.960	0.95	336.2	336.2	110.5	motor η (IE3), VFD η 0.97, input pf 0.95 (VFD front end), cable length
P-101	Rich amine pump (T-101 sump → E-101 → T-102 top)	continuous	400	DOL	37.8	45	0.942	0.85	40.1	40.1	24.8	Δz, ΔP exchanger/valve/line, pump η, cable length
P-102	Lean amine pump (E-102/T-102 sump → E-101 → E-104 → FV-201 → T-101 top)	continuous	400	DOL	39.9	55	0.946	0.87	42.1	42.1	23.9	Δz, ΔP exchanger/valve/line, pump η, cable length
P-103	Reflux pump (V-101 → FV-303 → T-102 top)	continuous	400	DOL	1.1	1.5	0.853	0.80	1.3	1.3	1.0	Δz, ΔP exchanger/valve/line, pump η, cable length
P-104	Wash water pump (wash sump → E-105 → wash distributor)	continuous	400	DOL	21.3	30	0.936	0.85	22.7	22.7	14.1	Δz, ΔP exchanger/valve/line, pump η, cable length
P-101B	Spare of P-101 (TBC: not on CC-PID-101)	standby	400	DOL	37.8	45	0.942	0.85	40.1	0.0	0.0	sparing philosophy TBC
P-102B	Spare of P-102 (TBC: not on CC-PID-101)	standby	400	DOL	39.9	55	0.946	0.87	42.1	0.0	0.0	sparing philosophy TBC
P-103B	Spare of P-103 (TBC: not on CC-PID-101)	standby	400	DOL	1.1	1.5	0.853	0.80	1.3	0.0	0.0	sparing philosophy TBC
P-104B	Spare of P-104 (TBC: not on CC-PID-101)	standby	400	DOL	21.3	30	0.936	0.85	22.7	0.0	0.0	sparing philosophy TBC
UPS-101	Instruments, DCS/SIS cabinets, analysers via UPS (allowance)	continuous	400	feeder	—	—	—	0.80	12.0	12.0	9.0	allowance
LTG	Area and platform lighting (allowance)	continuous	400	feeder	—	—	—	0.80	10.0	10.0	7.5	allowance
HT	Electric heat tracing: analyser sample lines, MEA make-up line (allowance)	continuous	400	feeder	—	—	—	1.00	5.0	5.0	0.0	allowance
SP	Small power, analyser-house HVAC (allowance)	continuous	400	feeder	—	—	—	0.85	15.0	15.0	9.3	allowance
P-105	Amine sump / drain pump (not shown, note 7; allowance)	intermittent	400	feeder	—	—	—	0.85	7.5	2.2	1.4	allowance
WO	Welding outlets (maintenance only)	standby	400	feeder	—	—	—	0.80	40.0	0.0	0.0	allowance

Input kW = shaft ÷ η_{motor} (÷ η_{VFD} for K-101); allowances are kVA × pf. Demand = input × demand factor of the duty. Spares carry no demand (standby).

3 · Maximum demand, transformer and MCC

Board	Demand kW	kvar	kVA	pf
LV MCC-101 (400 V)	150.4	90.9	175.8	0.856
MV feeder to K-101 VFD (6.6 kV)	336.2	110.5	353.9	0.950
Unit total	486.7	201.4	526.7	0.924

Transformer TR-101 6.6/0.4 kV: S ≥ 175.8 kVA × 1.25 = 219.8 kVA → **250 kVA** (next standard size)
Full-load current I = S/(√3·V) = 250 kVA/(√3 × 0.4 kV) = 361 A → MCC-101 main bus **400 A**
Prospective fault current at the LV bus ≈ I/u_k = 361/0.04 = **9.0 kA** (infinite MV source, conservative) → bus and breakers rated ≥ 25 kA (typical)
K-101 MV feeder current ≈ 354 kVA/(√3 × 6.6 kV) = 31 A

4 · Cable schedule (preliminary)

$I = P_{motor}/(\sqrt{3}\cdot V\cdot pf\cdot \eta)$; $I_{z,required} = I/(k_{ambient}\cdot k_{group})$; $\Delta V\ \% = \sqrt{3}\cdot I\cdot L\cdot(R\cos\varphi + X\sin\varphi)/V \times 100$, $R = \rho_{90°C}/A$, $X = 0.08\ \Omega/km$; DOL start: I × 6.5 at cos φ 0.3

Tag	V	Rating	Length m (assumed)	I A	Iz required A	Cable	Iz derated A	ΔV run %	ΔV start %
K-101	6600	355 kW	150	34.1	46.8	MV: 3C 6/10 kV XLPE, size by short-circuit withstand (TBC)	—	—	—
P-101	400	45 kW	90	81.1	111.4	3C 25 mm² Cu/XLPE	92.5	2.50	6.99
P-102	400	55 kW	80	96.5	132.5	3C 35 mm² Cu/XLPE	115.0	1.96	5.75
P-103	400	1.5 kW	70	3.2	4.4	3C 1.5 mm² Cu/XLPE	16.7	1.13	2.80
P-104	400	30 kW	110	54.4	74.8	3C 10 mm² Cu/XLPE	54.6	4.96	12.41
P-101B	400	45 kW	95	81.1	111.4	3C 25 mm² Cu/XLPE	92.5	2.64	7.38
P-102B	400	55 kW	85	96.5	132.5	3C 35 mm² Cu/XLPE	115.0	2.08	6.11
P-103B	400	1.5 kW	75	3.2	4.4	3C 1.5 mm² Cu/XLPE	16.7	1.21	3.00
P-104B	400	30 kW	115	54.4	74.8	3C 16 mm² Cu/XLPE	72.8	3.28	8.61
UPS-101	400	15 kVA	40	21.7	29.7	3C 2.5 mm² Cu/XLPE	23.3	2.66	2.66
LTG	400	12.5 kVA	60	18.0	24.8	3C 2.5 mm² Cu/XLPE	23.3	3.32	3.32
HT	400	5 kVA	60	7.2	9.9	3C 1.5 mm² Cu/XLPE	16.7	2.75	2.75
SP	400	17.6 kVA	50	25.4	34.9	3C 4 mm² Cu/XLPE	30.6	2.59	2.59
P-105	400	8.8 kVA	80	12.7	17.4	3C 2.5 mm² Cu/XLPE	23.3	3.31	3.31
WO	400	50 kVA	60	72.2	99.1	3C 16 mm² Cu/XLPE	72.8	2.15	2.15

Sizes are the smallest in the ampacity table that carry the derated current and meet both voltage-drop limits. Short-circuit withstand, earth-fault loop impedance and the MV cable are to be checked in detailed design (TBC).

5 · Hazardous-area classification note

Monoethanolamine is a combustible liquid: flash point ≈ 96 °C (pure MEA), auto-ignition ≈ 410 °C (typical SDS value, TBC). In this unit it is handled as a 30 wt% aqueous solution at up to 120 °C (TIC-206), i.e. above the flash point of pure MEA. Under IEC 60079-10-1 a liquid released above its flash point (or as a mist) can form an explosive atmosphere, so the hot-amine section must be assessed. CO2 is not flammable (it is an asphyxiant: gas detection and ventilation, not area classification). The flue gas carries no fuel.

Location	Fluid	Max. temperature (model)	Proposed classification (TBC)	Reason
T-101 sump, P-101 suction	Rich 30 wt% MEA	43.6 °C (TI108)	Assess: likely non-hazardous	handled 52 K below the flash point
E-101 rich outlet → T-102 feed	Rich 30 wt% MEA	109.0 °C (TI204)	Assess: Zone 2 IIA T3 proposed around flanges, seals, sample points	handled 13 K above the flash point (secondary grade releases)
T-102 bottom, E-102 kettle, P-102	Lean 30 wt% MEA	120.0 °C (TIC206)	Assess: Zone 2 IIA T3 proposed around flanges, seals, sample points	handled 24 K above the flash point (secondary grade releases)
E-101 lean outlet → E-104	Lean 30 wt% MEA	51.6 °C (TI205)	Assess: likely non-hazardous	handled 44 K below the flash point
T-102 overhead, E-103, V-101, P-103	CO2 + water vapour / water	40.0 °C (TIC302)	Non-hazardous (no flammable fluid)	CO2 / water / flue gas / steam
Flue gas duct, K-101, T-101 top	Flue gas (4 % CO2, 12 % O2, N2)	40.0 °C (TI101)	Non-hazardous (no flammable fluid)	CO2 / water / flue gas / steam

Location	Fluid	Max. temperature (model)	Proposed classification (TBC)	Reason
E-102 steam side	LP steam / condensate	138.9 °C (Ts)	Non-hazardous (no flammable fluid)	CO2 / water / flue gas / steam

Equipment in a classified area: gas group IIA, temperature class T3 (≤ 200 °C surface), which covers the hottest process surface (LP steam 138.9 °C) and stays well below the auto-ignition temperature. The flash point of the 30 wt% solution is to be tested; if it has no flash point at the handling temperature, the hot section may be declassified (EQ-EE-04).

6 · Engineering queries

No.	Query	Documents	Suggested resolution
EQ-EE-01	K-101 motor 355 kW (design shaft 313 kW × 1.10) is below the model's blower power at 100 % VFD speed (589 kW at 223 kNm³/h).	Load list MEA-EE-LL-001; P&ID; CC-PID-101 equipment list (313 kW); model FIC-101 range 0–250 kNm³/h, blower curve	Either cap the VFD speed (and FIC-101 output) so the motor cannot be overloaded, or size the motor for the full curve (next standard 630 kW).
EQ-EE-02	No spare pumps on the P&ID; the load list carries P-101B…P-104B as TBC standby (no demand).	CC-PID-101 equipment list vs load list	Confirm the sparing philosophy (2 × 100 % is typical for amine circulation).
EQ-EE-03	Pump differential pressures are not in any issued document; the model has no pump curves. The load list builds them from model pressures plus assumed elevations and equipment ΔP.	Load list vs HMB / Process engineering	Replace with the pump hydraulic calculation from the Process engineering tab when issued.
EQ-EE-04	Lean MEA is handled at 120 °C (TIC-206), above the 96 °C flash point of pure MEA; the flash point of the 30 wt% aqueous solution is not known.	Hazardous-area note in this load list; HAZOP; SDS	Test the solution flash point; finish the IEC 60079-10-1 assessment before the motor and instrument Ex ratings are frozen.
EQ-EE-05	Wash-water circulation (P-104) is not modelled; 300 m³/h is assumed from the E-105 duty on the sheet.	Model vs CC-PID-101 (E-105 5.8 MW) vs load list	Set the wash flow in the HMB.
EQ-EE-06	The model has a cooling-water supply switch (CWP) but CW pumps are on neither the P&ID; nor this load list.	Model (sw.CWP) vs CC-PID-101 vs load list	Define the battery limit: CW pumps by the host utility (assumed) or in this unit.